

Proposition 6.—Let $f^{(i)} \in \mathbb{Z}_p[X_1, \dots, X_m]$ be homogeneous polynomials with p -adic integer coefficients. The following are equivalent:

- a) The $f^{(i)}$ have a non trivial common zero in $(\mathbb{Q}_p)^m$.
- b) The $f^{(i)}$ have a common primitive zero in $(\mathbb{Z}_p)^m$.
- c) For all $n > 1$, the $f_n^{(i)}$ have a common primitive zero in $(A_n)^m$.

The implication b) $\Rightarrow$ a) is trivial. Conversely, if $x = (x_1, \dots, x_m)$ is a nontrivial common zero of the $f^{(i)}$, put

$$h = \inf(v_p(x_1), \dots, v_p(x_m)) \quad \text{and} \quad y = p^{-h}x.$$

It is clear that y is a primitive element of $(\mathbb{Z}_p)^m$, and that it is a common zero of the $f^{(i)}$. Hence b) $\Leftrightarrow$ a).

The equivalence of b) and c) follows from the above lemma.

2.2. Amelioration of approximate solutions.

We are concerned with passing from a solution (mod p^n) to a true solution (i.e. with coefficients in $\mathbb{Z}_p$). One uses the following lemma (p -adic analogue of “Newton’s method”):

Lemma.—Let $f \in \mathbb{Z}_p[X]$ and let f' be its derivative. Let $x \in \mathbb{Z}_p$, $n, k \in \mathbb{Z}$ such that $0 \leq 2k < n$, $f(x) \equiv 0 \pmod{p^n}$, $v_p(f'(x)) = k$. Then there exists $y \in \mathbb{Z}_p$ such that

$$f(y) \equiv 0 \pmod{p^{n+1}}, \quad v_p(f'(y)) = k, \quad \text{and} \quad y \equiv x \pmod{p^{n-k}}.$$

Take y of the form $x + p^{n-k}z$ with $z \in \mathbb{Z}_p$. By Taylor’s formula we have

$$f(y) = f(x) + p^{n-k}zf'(x) + p^{2n-2k}a \quad \text{with } a \in \mathbb{Z}_p.$$

By hypothesis $f(x) = p^n b$ and $f'(x) = p^k c$ with $b \in \mathbb{Z}_p$ and $c \in \mathbb{U}$. This allows us to choose z in such a way that

$$b + zc \equiv 0 \pmod{p}.$$

From this we get

$$f(y) = p^n(b + zc) + p^{2n-2k}a \equiv 0 \pmod{p^{n+1}}$$

since $2n - 2k > n$. Finally Taylor’s formula applied to f' shows that $f'(y) \equiv p^k c \pmod{p^{n-k}}$; since $n - k > k$, we see that $v_p(f'(y)) = k$.

Theorem 1.—Let $f \in \mathbb{Z}_p[X_1, \dots, X_m]$, $x = (x_i) \in (\mathbb{Z}_p)^m$, $n, k \in \mathbb{Z}$ and j an integer such that $0 \leq j \leq m$. Suppose that $0 < 2k < n$ and that

$$f(x) \equiv 0 \pmod{p^n} \quad \text{and} \quad v_p\left(\frac{\partial f}{\partial X_j}(x)\right) = k.$$

Then there exists a zero y of f in $(\mathbb{Z}_p)^m$ which is congruent to x modulo p^{n-k} .

Suppose first that $m = 1$. By applying the above lemma to $x^{(0)} = x$, we obtain $x^{(1)} \in \mathbb{Z}_p$ congruent to $x^{(0)} \pmod{p^{n-k}}$ and such that

$$f(x^{(1)}) \equiv 0 \pmod{p^{n+1}} \quad \text{and} \quad v_p(f'(x^{(1)})) = k.$$

We can apply the lemma to $x^{(1)}$, after replacing n by $n+1$. Arguing inductively, we construct in this way a sequence $x^{(0)}, \dots, x^{(q)}, \dots$ such that

$$x^{(q+1)} \equiv x^{(q)} \pmod{p^{n+q-k}}, \quad f(x^{(q)}) \equiv 0 \pmod{p^{n+q}}.$$

This is a Cauchy sequence. If y is its limit, we have $f(y) = 0$ and $y \equiv x \pmod{p^{n-k}}$, hence the theorem for $m = 1$.

The case $m > 1$ reduces to the case $m = 1$ by modifying only x_j . More precisely, let $\tilde{f} \in \mathbb{Z}_p[X_j]$ be the polynomial in one variable obtained by replacing X_i , $i \neq j$, by x_i . What has just been proven can be applied to $\tilde{f}$ and x_j ; this shows the existence of $y_j \equiv x_j \pmod{p^{n-k}}$ such that $\tilde{f}(y_j) = 0$. If one puts $y_i = x_i$ for $i \neq j$, the element $y = (y_i)$ satisfies the desired condition.

Corollary 1.—Every simple zero of the reduction modulo p of a polynomial f lifts to a zero of f with coefficients in $\mathbb{Z}_p$.

(If g is a polynomial over a field k , a zero x of g is called *simple* if at least one of the partial derivatives $\partial g / \partial X_j$ is nonzero at x .)

This is the special case $n = 1$, $k = 0$.

Corollary 2.—Suppose $p \neq 2$. Let $f(X) = \sum a_{ij} X_i X_j$ with $a_{ij} = a_{ji}$ be a quadratic form with coefficients in $\mathbb{Z}_p$ whose discriminant $\det(a_{ij})$ is invertible. Let $a \in \mathbb{Z}_p$. Every primitive solution of the equation $f(x) \equiv a \pmod{p}$ lifts to a true solution.

In view of cor. 1, it suffices to show that x does not annihilate all the partial derivatives of f modulo p . Now $\frac{\partial f}{\partial X_i} = 2 \sum_j a_{ij} X_j$; since $\det(a_{ij}) \not\equiv 0 \pmod{p}$ and x is primitive, one of these partial derivatives is $\not\equiv 0 \pmod{p}$.

Corollary 3.—Suppose $p = 2$. Let $f = \sum a_{ij} X_i X_j$ with $a_{ij} = a_{ji}$ be a quadratic form with coefficients in $\mathbb{Z}_2$ and let $a \in \mathbb{Z}_2$. Let x be a primitive solution of $f(x) \equiv a \pmod{8}$. We can lift x to a true solution provided x does not annihilate all the $\frac{\partial f}{\partial X_j}$ modulo 4; this last condition is fulfilled if $\det(a_{ij})$ is invertible.

The first assertion follows from the theorem applied to $n = 3$, $k = 1$; the second can be proved as in the case $p \neq 2$ (taking into account the factor 2).

§3. The multiplicative group of $\mathbb{Q}_p$

3.1. The filtration of the group of units

Let $U = \mathbb{Z}_p^*$ be the group of p -adic units. For every $n \geq 1$, put $U_n = 1 + p^n \mathbb{Z}_p$; this is the kernel of the homomorphism $\epsilon_n: U \rightarrow (\mathbb{Z}/p^n \mathbb{Z})^*$. In particular, the quotient U/U_1 can be identified with $\mathbb{F}_p^*$, hence is cyclic of order $p-1$ (cf. Chap. I, th. 2). The U_n form a decreasing sequence of open subgroups of U , and $U = \varprojlim U/U_n$. If $n \geq 1$, the map

$$(1 + p^n x) \mapsto (x \text{ modulo } p)$$